

GNU nano 8.7.1

task2.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("Enter command (or exit): ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = 0;

        if(strcmp(command, "exit")==0)
            break;

        pid_t pid = fork();

        if(pid < 0)
        {
            printf("Fork Failed\n");
        }
        else if(pid == 0)
        {
            execlp(command, command, NULL);

            printf("Invalid Command\n");
            exit(1);
        }
        else
        {
            wait(NULL);
        }
    }

    return 0;
}
```

shreyaj@LAPTOP-FT6VVM4D:~/OSSP\$ tree

```
├── FileManagerment
│   ├── b.txt
│   ├── file1.txt
│   ├── hello
│   ├── newcopy.txt
│   └── numbers.txt
├── LICENSE
├── Makefile
├── ProcessTask
│   └── task2
│       └── task2.c
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── builtin.h
│   ├── design.md
│   ├── executor.h
│   ├── parser.h
│   ├── report.pdf
│   └── shell.h
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
├── structure.txt
└── tests
    ├── test_executor.c
    └── test_parser.c
```

11 directories, 30 files

shreyaj@LAPTOP-FT6VVM4D:~/OSSP\$

```
shreyaj@LAPTOP-FT6VVM4D: ~$ mkdir ProcessTask
mkdir: ProcessTask: File exists
shreyaj@LAPTOP-FT6VVM4D: ~$ cd ProcessTask
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ nano task2.c
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ nano task2.c
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ gcc task2.c -o task2
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ sudo apt update
sudo apt install gcc
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.4 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Hit:6 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Fetched 1118 kB in 5s (208 kB/s)
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
gcc is already the newest version (4:15.2.0-5ubuntu1).
gcc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 99
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ ./task2
Enter command (or exit): date
Wed Aug  5 09:29:00 UTC 2026
Enter command (or exit): pwd
/home/shreyaj/ProcessTask
Enter command (or exit): ls
task2  task2.c
Enter command (or exit): whoami
shreyaj
Enter command (or exit): exit
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$
```

```

shreyaj@LAPTOP-FT6VVM4D:~$ uname -a
Linux LAPTOP-FT6VVM4D 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
IC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
shreyaj@LAPTOP-FT6VVM4D:~$ ls
Documents  OS_LAB          file.txt          top_output.txt
First      ProcessTask      lscpu_output.txt  uname_output.txt
First.c    abc.txt          process.c.save
OSSP       assignment1_history.txt  ps_output.txt
shreyaj@LAPTOP-FT6VVM4D:~$ ls -l
total 56
drwxr-xr-x  3 shreyaj shreyaj 4096 Aug  5 09:02 Documents
-rwxr-xr-x  1 shreyaj shreyaj 15768 Aug  4 17:24 First
-rw-r--r--  1 shreyaj shreyaj  23 Aug  4 17:24 First.c
drwxr-xr-x 13 shreyaj shreyaj 4096 Aug  4 17:09 OSSP
drwxr-xr-x  2 shreyaj shreyaj 4096 Aug  4 18:46 OS_LAB
drwxr-xr-x  2 shreyaj shreyaj 4096 Aug  5 09:27 ProcessTask
-rw-r--r--  1 shreyaj shreyaj  0 Aug  4 17:24 abc.txt
-rw-r--r--  1 shreyaj shreyaj 3231 Aug  4 18:34 assignment1_history.txt
-rw-r--r--  1 shreyaj shreyaj  0 Aug  5 09:01 file.txt
-rw-r--r--  1 shreyaj shreyaj 3211 Aug  4 18:50 lscpu_output.txt
-rw-r--r--  1 shreyaj shreyaj  0 Aug  4 17:24 process.c.save
-rw-r--r--  1 shreyaj shreyaj 2869 Aug  4 18:50 ps_output.txt
-rw-r--r--  1 shreyaj shreyaj 3210 Aug  4 18:50 top_output.txt
-rw-r--r--  1 shreyaj shreyaj  125 Aug  4 18:50 uname_output.txt
shreyaj@LAPTOP-FT6VVM4D:~$ ls -a
.      .landscape  First.c          lscpu_output.txt
..     .lesshtst   OSSP             process.c.save
.bash_history .local      OS_LAB           ps_output.txt
.bash_logout .motd_shown ProcessTask       top_output.txt
.bashrc      .profile    abc.txt          uname_output.txt
.cache       Documents   assignment1_history.txt
.config      First       file.txt
shreyaj@LAPTOP-FT6VVM4D:~$ cd
shreyaj@LAPTOP-FT6VVM4D:~$ cd ..
shreyaj@LAPTOP-FT6VVM4D:/home$ mkdir lab
mkdir: Permission denied
shreyaj@LAPTOP-FT6VVM4D:/home$ rmdir
error: the following required arguments were not provided:
  <dirs>...

Usage: rmdir [OPTION]... DIRECTORY...

For more information, try '--help'.
shreyaj@LAPTOP-FT6VVM4D:/home$ rmdir test

```

```
shreyaj@LAPTOP-FT6VVM4D: ~$ pwd
/home/shreyaj
shreyaj@LAPTOP-FT6VVM4D: ~$ history
1  sudo apt update
2  sudo apt install build-essential
3  gcc -version
4  gcc --version
5  pwd
6  ls
7  is
8  cd OSSP
9  mkdir OSSP
10 pwd
11 mkdir OSSP
12 cd OSSP
13 pwd
14 mkdir src include obj bin docs tests scripts assets
15 mkdir src include obj bin docs tests scripts assets
16 ls
17 cd src
18 pwd
19 touch main.c shell.c parser.c executor.c builtin.c
20 ls
21 cd ..
22 cd include
23 touch shell.h parser.h executor.h builtin.h
24 ls
25 cd ..
26 cd docs
27 touch shell.h parser.h executor.h builtin.h
28 pwd
29 mkdir OSSP
30 cd OSSP
31 pwd
32 mkdir include obj bin docs tests scripts assets
33 ls
34 cd src
35 touch main.c shell.c parser.c executor.c builtin.c
36 ls
37 cd ..
38 cd include
39 touch shell.h parser.h executor.h builtin.h
40 ls
41 cd ..
42 cd docs
```

```
41 cd ..
42 cd docs
43 touch design.md report.pdf
44 cd ..
45 cd tests
46 touch test_parser.c test_executor.c
47 cd ..
48 cd scripts
49 touch run.sh
50 cd ..
51 cd assets
52 touch architecture.png
53 cd ..
54 touch README.md LICENSE Makefile .gitignore
55 ls -a
56 git init
57 ls -a
58 sudo apt install tree
59 tree
60 tree > structure.txt
61 gcc -version
62 gcc --version
63 git --version
64 cd
65 touch file.txt
66 cd ~/Documents
67 gcc --version
68 git --version
69 cd
70 touch file.txt
71 cd ~/Documents
72 ls ~
73 history
74 gcc --version
75 git --version
76 cd
77 touch file.txt
78 cd ~/OSSP
79 mkdir FileManagement
80 cd FileManagement
81 touch file1.txt
82 ls file1.txt
83 touch a.txt b.txt c.txt
84 echo "Hello Ubuntu"> file1.txt
85 cat file1.txt
```

```
86 cat file.txt a.txt
87 cat file1.txt a.txt
88 cp file1.txt copy.txt
89 ls a.txt b.txt c.txt copy.txt file1.txt
90 mv copy.txt newcopy.txt
91 ls
92 rm a.txt c.txt
93 seq 20>numbers.txt
94 seq 20 > numbers.txt
95 head numbers.txt
96 tail numbers.txt
97 nano hello
98 ls
99 ./hello
100 which gcc
101 whereis gcc
102 man gcc
103 ls -la gcc_manual.txt
104 ls
105 explorer.exe .
106 pwd
107 history
108 whoami
109 date
110 cal
111 hostname
112 uname -a
113 ls
114 ls -l
115 ls -a
116 cd ~
117 mkdir -p OSSP/ProcessTask
118 cd ~/OSSP/ProcessTask
119 nanotask2.c
120 nano task2.c
121 gcc task2.c -o task2
122 nano task2.c
123 gcc task2.c -o task2
124 ./task2
125 cd~
126 cd ~
127 pwd
128 cd Desktop
129 cd ~
130 mkdir ProcessTask
```

```
131 cd ProcessTask
132 nano task2.c
133 gcc task2.c -o task2
134 ./task2
135 pwd
136 history
137 whoami
138 date
139 cal
140 hostname
141 uname -a
142 ls
143 cd ~
144 mkdir Documents
145 touch First.c
146 nano First.c
147 gcc First.c -o First
148 touch abc.txt process.c.save
149 ls
150 ls -a
151 ls -l
152 cd
153 cd ..
154 mkdir lab
155 rmdir
156 rmdir test
157 rmdir lab
158 find
159 find . -name "*.OSSP"
160 cd ~
161 history
162 explore.exe
163 explorer.exe
164 history > assignment1_history.txt
165 explorer.exe
166 history > assignment1_history.txt
167 explorer.exe .
168 pwd
169 ls
170 mkdir OS_LAB
171 cd OS_LAB
172 nano task1.c
173 gcc task1.c -o task1
174 sudo apt update
175 sudo apt install build-essential
```

```
shreyaj@LAPTOP-FT6VVM4D: × Ubuntu × Ubuntu × Ubuntu × + v - ×
175 sudo apt install build-essential
176 ./task1
177 uname -a
178 lscpu
179 ps
180 top
181 uname -a
182 lscpu
183 ps
184 top
185 uname -a > uname_output.txt
186 lscpu > lscpu_output.txt
187 ps -ef > ps_output.txt
188 top -b -n 1 > top_output.txt
189 ls
190 histor
191 history
192 pwd
193 history
shreyaj@LAPTOP-FT6VVM4D:~$ whoami
shreyaj
shreyaj@LAPTOP-FT6VVM4D:~$ date
Wed Aug 5 09:32:58 UTC 2026
shreyaj@LAPTOP-FT6VVM4D:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
shreyaj@LAPTOP-FT6VVM4D:~$ hostname
LAPTOP-FT6VVM4D
shreyaj@LAPTOP-FT6VVM4D:~$ uname -a
Linux LAPTOP-FT6VVM4D 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
shreyaj@LAPTOP-FT6VVM4D:~$ ls
Documents  OS_LAB          file.txt        top_output.txt
First      ProcessTask     lscpu_output.txt  uname_output.txt
First.c    abc.txt         process.c.save    ps_output.txt
OSSP       assignment1_history.txt
shreyaj@LAPTOP-FT6VVM4D:~$ ls -l
total 56
drwxr-xr-x  3 shreyaj shreyaj 4096 Aug  5 09:02 Documents
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-rw-r--r--  1 shreyaj shreyaj   23 Aug  4 17:24 First.c
drwxr-xr-x 13 shreyaj shreyaj 4096 Aug  4 17:09 OSSP
drwxr-xr-x  2 shreyaj shreyaj 4096 Aug  4 18:46 OS_LAB
drwxr-xr-x  2 shreyaj shreyaj 4096 Aug  5 09:27 ProcessTask
-rw-r--r--  1 shreyaj shreyaj   0 Aug  4 17:24 abc.txt
-rw-r--r--  1 shreyaj shreyaj 3231 Aug  4 18:34 assignment1_history.txt
```

```
shreyaj@LAPTOP-FT6VVM4D: x  Ubuntu x  Ubuntu x  Ubuntu x  +  v  -  [  X
-rw-r--r-- 1 shreyaj shreyaj  0 Aug  5 09:01 file.txt
-rw-r--r-- 1 shreyaj shreyaj 3211 Aug  4 18:50 lscpu_output.txt
-rw-r--r-- 1 shreyaj shreyaj  0 Aug  4 17:24 process.c.save
-rw-r--r-- 1 shreyaj shreyaj 2869 Aug  4 18:50 ps_output.txt
-rw-r--r-- 1 shreyaj shreyaj 3210 Aug  4 18:50 top_output.txt
-rw-r--r-- 1 shreyaj shreyaj  125 Aug  4 18:50 uname_output.txt
shreyaj@LAPTOP-FT6VVM4D:~$ ls -la
.          .landscape  First.c          lscpu_output.txt
..         .lessht     OSSP            process.c.save
.bash_history .local      OS_LAB          ps_output.txt
.bash_logout .motd_shown ProcessTask      top_output.txt
.bashrc      .profile    abc.txt         uname_output.txt
.cache       Documents   assignment1_history.txt
.config      First       file.txt
shreyaj@LAPTOP-FT6VVM4D:~$ cd
shreyaj@LAPTOP-FT6VVM4D:~$ cd ..
shreyaj@LAPTOP-FT6VVM4D:~/home$ mkdir lab
mkdir: Permission denied
shreyaj@LAPTOP-FT6VVM4D:~/home$ rmdir
error: the following required arguments were not provided:
  <dirs>...

Usage: rmdir [OPTION]... DIRECTORY...

For more information, try '--help'.
shreyaj@LAPTOP-FT6VVM4D:~/home$ cd ~
mkdir lab
shreyaj@LAPTOP-FT6VVM4D:~$ rmdir folder_name
rmdir: failed to remove 'folder_name': No such file or directory
shreyaj@LAPTOP-FT6VVM4D:~$ rmdir test
rmdir: failed to remove 'test': No such file or directory
shreyaj@LAPTOP-FT6VVM4D:~$ rmdir lab
shreyaj@LAPTOP-FT6VVM4D:~$ find
.
./assignment1_history.txt
./OS_LAB
./OS_LAB/task1
./OS_LAB/task1.c
./.local
./.local/share
./.local/share/nano
./process.c.save
./ps_output.txt
./.motd_shown
./OSSP
```

```
./OSSP/docs
./OSSP/docs/design.md
./OSSP/docs/report.pdf
./OSSP/docs/shell.h
./OSSP/docs/parser.h
./OSSP/docs/builtin.h
./OSSP/docs/executor.h
./OSSP/assets
./OSSP/assets/architecture.png
./OSSP/Makefile
./OSSP/LICENSE
./OSSP/FileManagement
./OSSP/FileManagement/b.txt
./OSSP/FileManagement/newcopy.txt
./OSSP/FileManagement/numbers.txt
./OSSP/FileManagement/hello
./OSSP/FileManagement/file1.txt
./OSSP/.gitignore
./OSSP/tests
./OSSP/tests/test_parser.c
./OSSP/tests/test_executor.c
./OSSP/structure.txt
./OSSP/src
./OSSP/src/main.c
./OSSP/src/shell.c
./OSSP/src/parser.c
./OSSP/src/builtin.c
./OSSP/src/executor.c
./OSSP/scripts
./OSSP/scripts/run.sh
./OSSP/README.md
./OSSP/include
./OSSP/include/shell.h
./OSSP/include/parser.h
./OSSP/include/builtin.h
./OSSP/include/executor.h
./OSSP/ProcessTask
./OSSP/ProcessTask/task2.c
./OSSP/ProcessTask/task2
./OSSP/bin
./OSSP/obj
./OSSP/.git
./OSSP/.git/hooks
./OSSP/.git/hooks/prepare-commit-msg.sample
./OSSP/.git/hooks/post-update.sample
```

```
./OSSP/.git/info
./OSSP/.git/info/exclude
./OSSP/.git/HEAD
./OSSP/.git/objects
./OSSP/.git/objects/info
./OSSP/.git/objects/pack
./OSSP/.git/config
./OSSP/.git/description
./OSSP/.git/refs
./OSSP/.git/refs/tags
./OSSP/.git/refs/heads
./top_output.txt
./First
./abc.txt
./bash_history
./ProcessTask
./ProcessTask/task2.c
./ProcessTask/task2
./Documents
./Documents/FileManagement
./Documents/FileManagement/b.txt
./Documents/FileManagement/newcopy.txt
./Documents/FileManagement/numbers.txt
./Documents/FileManagement/hello
./Documents/FileManagement/file1.txt
./bashrc
./lscpu_output.txt
./landscap
./landscap/sysinfo.log
./First.c
./bash_logout
./config
./config/ubuntu-insights
./config/ubuntu-insights/linux-consent.toml
./config/ubuntu-insights/wsl_setup-consent.toml
./config/ubuntu-insights/ubuntu_release_upgrader-consent.toml
./config/procps
./cache
./cache/ubuntu-insights
./cache/ubuntu-insights/wsl_setup
./cache/ubuntu-insights/wsl_setup/local
./cache/ubuntu-insights/wsl_setup/uploaded
./cache/ubuntu-insights/wsl_setup/uploaded/1785130289.json
./cache/ubuntu-insights/linux
./cache/ubuntu-insights/linux/local
```